CHAPTER 3

METHODS FOR SOLVING SINUSOIDAL CIRCUITS

3.1 EQUIVALENT TRANSFORMATION METHOD

❑ General method:

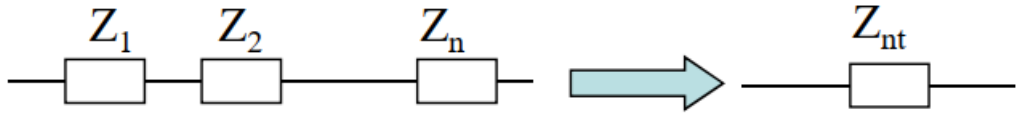
Step 1: Convert to complex form

- Voltage source $u(t) = U\sqrt{2}\sin(\omega t + \varphi)$ becomes $\dot{\mathbf{U}} = U\angle\varphi$
- R, L, and C become $\boldsymbol{Z}_R = R$, $\boldsymbol{Z}_L = j\omega L$, $\boldsymbol{Z}_C = \dfrac{1}{j\omega C}$

Step 2: Solve the complex circuit by the methods in this chapter, e.g. find current $\dot{\mathbf{I}}$

Step 3: Put $\dot{\mathbf{I}}$ in polar form $I\angle\alpha$ and deduce the actual current: $i(t) = I\sqrt{2}\sin(\omega t + \alpha)$

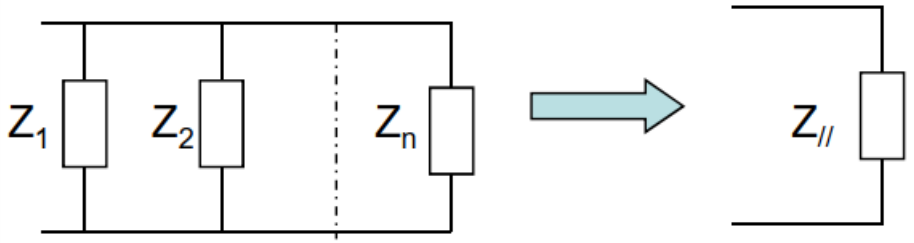
❑ Series impedance



$$\mathbf{Z}_{td} = \mathbf{Z}_{nt} = \sum_{k=1}^{n}\mathbf{Z}_k = \sum_{k=1}^{n}R_k + j\sum_{k=1}^{n}X_k \qquad (3.1)$$

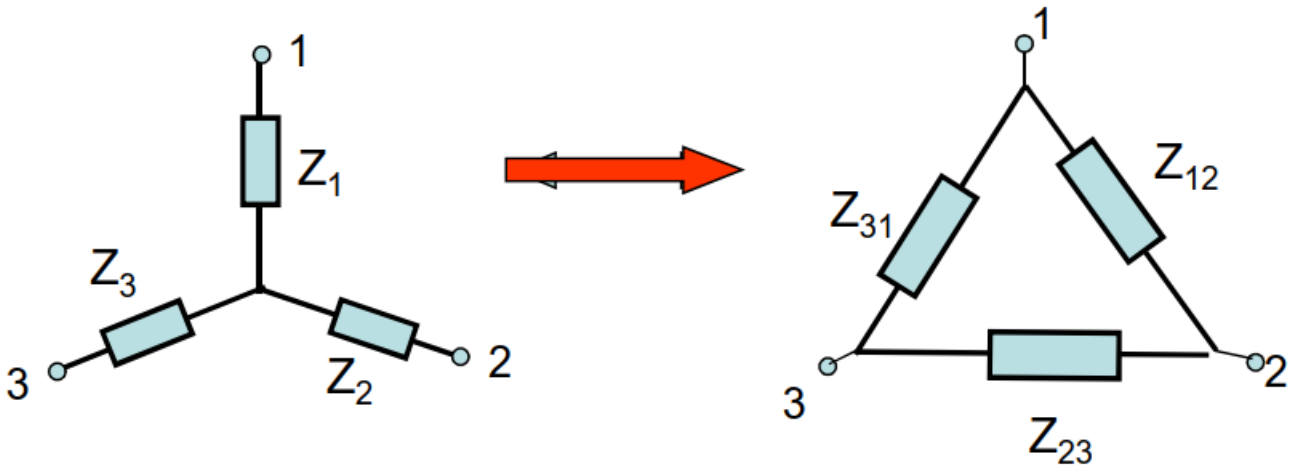
3.1 EQUIVALENT TRANSFORMATION METHOD

❑ Parallel impedance



$$\mathbf{Z}_{td} = \mathbf{Z}_{//} = \frac{1}{\sum_{k=1}^{n} \frac{1}{\mathbf{Z}_k}} \tag{3.2}$$

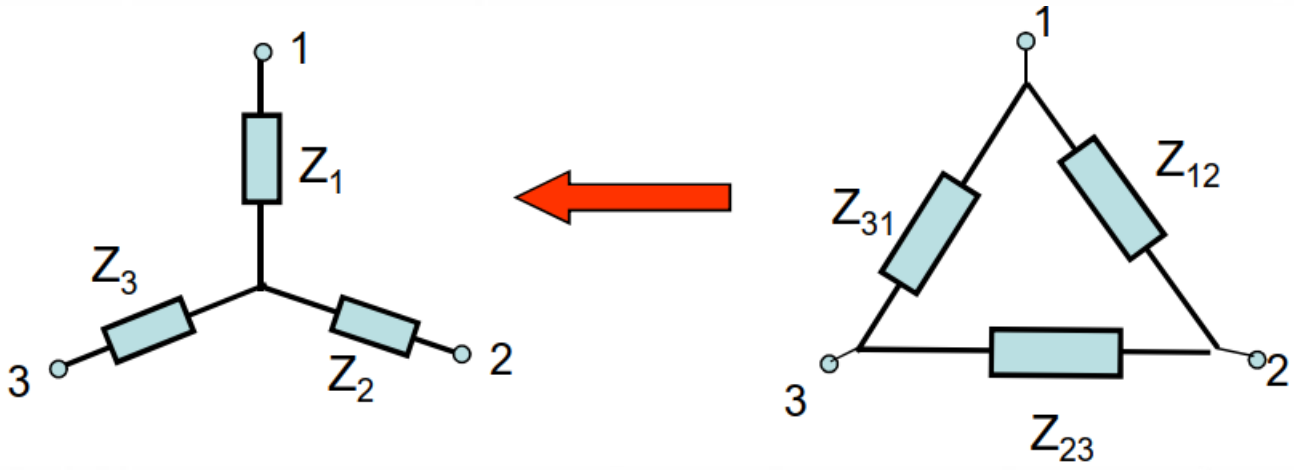
❑ Star (Y) – delta (Δ)



$$\begin{cases} \mathbf{Z}_{12} = \mathbf{Z}_1 + \mathbf{Z}_2 + \dfrac{\mathbf{Z}_1\mathbf{Z}_2}{\mathbf{Z}_3} \\ \mathbf{Z}_{23} = \mathbf{Z}_2 + \mathbf{Z}_3 + \dfrac{\mathbf{Z}_2\mathbf{Z}_3}{\mathbf{Z}_1} \\ \mathbf{Z}_{31} = \mathbf{Z}_3 + \mathbf{Z}_1 + \dfrac{\mathbf{Z}_3\mathbf{Z}_1}{\mathbf{Z}_2} \end{cases} \tag{3.3}$$

3.1 EQUIVALENT TRANSFORMATION METHOD

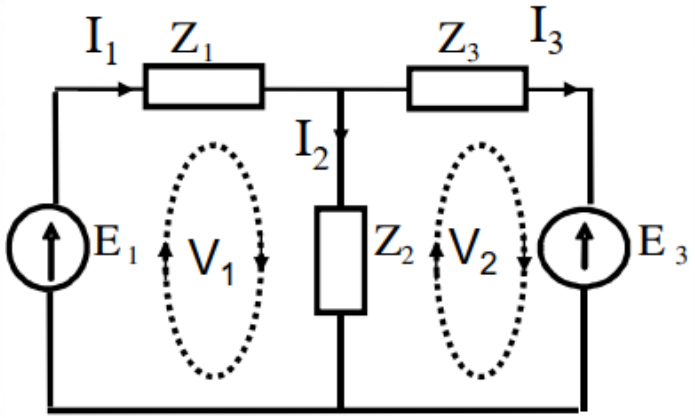
❑ Delta (Δ) – Star (Y)



$$\begin{cases} \mathbf{Z}_1 = \dfrac{\mathbf{Z}_{12}\mathbf{Z}_{31}}{\mathbf{Z}_{12}+\mathbf{Z}_{23}+\mathbf{Z}_{31}} \\ \mathbf{Z}_2 = \dfrac{\mathbf{Z}_{12}\mathbf{Z}_{23}}{\mathbf{Z}_{12}+\mathbf{Z}_{23}+\mathbf{Z}_{31}} \\ \mathbf{Z}_3 = \dfrac{\mathbf{Z}_{23}\mathbf{Z}_{31}}{\mathbf{Z}_{12}+\mathbf{Z}_{23}+\mathbf{Z}_{31}} \end{cases} \quad (3.4)$$

❑ Exercises

3.2 BRANCH CURRENT METHOD



Step 1: The circuit has 2 nodes, 3 branches. Choose 3 branch currents $\boldsymbol{I_1, I_2, I_3}$ as unknowns (directions chosen arbitrarily)

Apply KCL at one node: $\mathbf{I}_1 = \mathbf{I}_2 + \mathbf{I}_3 \Rightarrow \mathbf{I}_2 = \mathbf{I}_1 - \mathbf{I}_3$ (3.5)

Step 2: Use Ohm's law to compute **U** from **I**

$$\mathbf{U}_1 = \mathbf{Z}_1\mathbf{I}_1,\ \mathbf{U}_2 = \mathbf{Z}_2\mathbf{I}_2,\ \mathbf{U}_3 = \mathbf{Z}_3\mathbf{I}_3$$

Step 3: Apply KVL to the 2 meshes:

- Mesh containing $\boldsymbol{I_1}$: $\mathbf{U}_1 + \mathbf{U}_2 = \mathbf{E}_1 \Rightarrow \mathbf{Z}_1\mathbf{I}_1 + \mathbf{Z}_2\mathbf{I}_2 = \mathbf{E}_1$ (3.6)
- Mesh containing $\boldsymbol{I_3}$: $\mathbf{U}_3 - \mathbf{U}_2 = -\mathbf{E}_3 \Rightarrow \mathbf{Z}_3\mathbf{I}_3 - \mathbf{Z}_2\mathbf{I}_2 = -\mathbf{E}_3$ (3.7)

Solving 3 equations (3.5), (3.6), and (3.7) gives I_1, I_2, I_3

3.3 MESH CURRENT METHOD

- Mesh: independent loops; number of meshes is calculated by:

$$k = n - m + 1 \tag{3.8}$$

where, n = branches; m = nodes

- If the circuit has k meshes, the equations for the k mesh currents $\boldsymbol{I}_1, .. \boldsymbol{I}_k$ are:

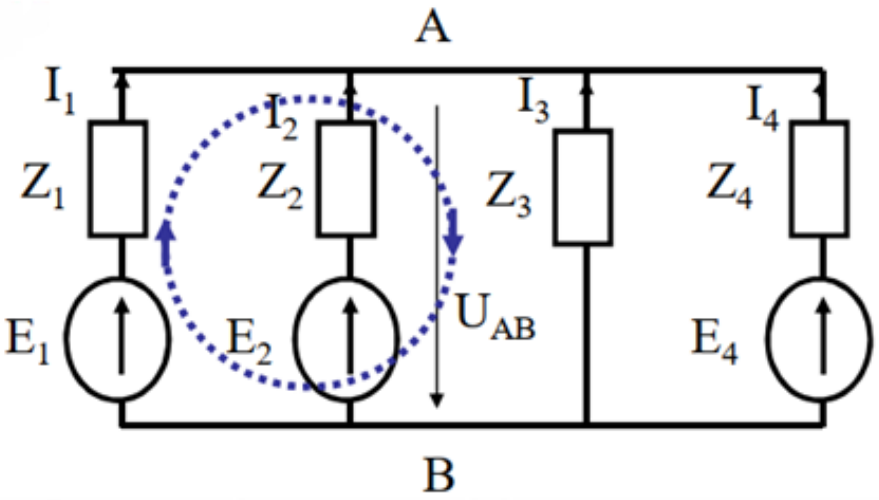
$$\begin{cases} +\mathbf{Z}_{11}\mathbf{I}_1 - \mathbf{Z}_{12}\mathbf{I}_2 - \ldots - \mathbf{Z}_{1k}\mathbf{I}_k = \mathbf{E}_1 \\ -\mathbf{Z}_{21}\mathbf{I}_1 + \mathbf{Z}_{22}\mathbf{I}_2 - \ldots - \mathbf{Z}_{2k}\mathbf{I}_k = \mathbf{E}_2 \\ \qquad\qquad \ldots\ldots\ldots\ldots \\ -\mathbf{Z}_{k1}\mathbf{I}_1 - \mathbf{Z}_{k2}\mathbf{I}_2 - \ldots + \mathbf{Z}_{kk}\mathbf{I}_k = \mathbf{E}_k \end{cases} \tag{3.9}$$

where, $\mathbf{Z}_{ii}$ is the total impedance of mesh i

$\mathbf{Z}_{ij} = \mathbf{Z}_{ji}$ is the impedance shared by meshes i and j

$\mathbf{E}_i$ is the algebraic sum of voltage sources in mesh i, taken along $\boldsymbol{I}_i$

3.4 TWO-NODE VOLTAGE METHOD



- Take the two-node voltage as the unknown: $\mathbf{U}_{AB}$
- Apply Kirchhoff's laws to set up equations for the two-node voltage
- Find the branch currents from the two-node voltage

Apply KCL at node A: $\sum_{k=1}^{4}\mathbf{I}_k = 0$

Hence: $\mathbf{Z}_1\mathbf{I}_1 + \mathbf{U}_{AB} = \mathbf{E}_1;\ \mathbf{Z}_2\mathbf{I}_2 + \mathbf{U}_{AB} = \mathbf{E}_2;\ \Rightarrow \mathbf{Z}_k\mathbf{I}_k + \mathbf{U}_{AB} = \mathbf{E}_k \Rightarrow \mathbf{I}_k = \dfrac{\mathbf{E}_k - \mathbf{U}_{AB}}{\mathbf{Z}_k}$

$$\Rightarrow \sum_{k=1}^{4}\frac{\mathbf{E}_k - \mathbf{U}_{AB}}{\mathbf{Z}_k} = 0$$

Let: $\dfrac{1}{\mathbf{Z}_k} = \mathbf{Y}_k \Rightarrow \sum_{k=1}^{4}\mathbf{Y}_k(\mathbf{E}_k - \mathbf{U}_{AB}) = 0 \Rightarrow \sum_{k=1}^{4}\mathbf{Y}_k\mathbf{E}_k = \mathbf{U}_{AB}\sum_{k=1}^{4}\mathbf{Y}_k$

$$\Rightarrow \mathbf{U}_{AB} = \frac{\sum_{k=1}^{4}\mathbf{Y}_k\mathbf{E}_k}{\sum_{k=1}^{4}\mathbf{Y}_k} \qquad (3.10)$$

3.5 SUPERPOSITION METHOD

- **Superposition theorem**: in a linear circuit with several sources, the current (or voltage) in a branch is the algebraic sum of those produced by each independent source acting alone; the other sources are turned off (voltage sources shorted, current sources opened).

Step 1: Let source 1 act alone, the others are off. Solve the circuit for the component current $\boldsymbol{I_1}$ of the current **I** to be found

Step 2: Repeat for the remaining sources to obtain the component currents $\boldsymbol{I_2}$, $\boldsymbol{I_3}$,…, $\boldsymbol{I_n}$ of the current **I** to be found

$$\mathbf{I} = \mathbf{I}_1 + \mathbf{I}_2 + \ldots + \mathbf{I}_n \tag{3.11}$$

Note: - The component currents must be drawn at the right place and direction as **I**

- If the circuit has a DC source ($\varpi = 0$) the inductor must be short-circuited $\boldsymbol{Z_L} = 0$ and the capacitor open-circuited $\boldsymbol{Z_C} = \infty$